

To display the first 5 records of the data set you can write this textual query:

```
response = pandas_ai(df, "Show 5 rows of data in tabular form")
print(response)
```

The output is shown in Figure 3.

A more detailed conversational query can be written as follows:

```
response = pandas_ai(df, "Show 15 rows of data in tabular form columns Year, DIST_NAME, RICE_PRODUCTION_1000_tons of DIST_NAME Hooghly RICE_PRODUCTION_1000_tons GREATER THAN 400")
print(response)
```

The output is shown in Figure 4.

Similarly, we can also execute the following query over the DataFrame.

	Year	Dist_Name	RICE_PRODUCTION_1000_tons
164	1974	Hooghly	400.21
165	1975	Hooghly	404.47
167	1977	Hooghly	432.73
170	1980	Hooghly	510.82
173	1983	Hooghly	475.21
174	1984	Hooghly	472.88
175	1985	Hooghly	479.15
176	1986	Hooghly	458.60
177	1987	Hooghly	508.95
178	1988	Hooghly	584.78
179	1989	Hooghly	565.28
180	1990	Hooghly	610.69

Figure 4: Result of a detailed conversational query

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 364 entries, 0 to 363
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Year                                364 non-null   int64
1   Dist_Name                           364 non-null   object
2   RICE_AREA_1000_ha                   364 non-null   float64
3   RICE_PRODUCTION_1000_tons           364 non-null   float64
4   RICE_YIELD_Kg_per_ha                 364 non-null   float64
dtypes: float64(3), int64(1), object(1)
memory usage: 14.3+ KB
```

Figure 5: Basic structures of a DataFrame

```
response = pandas_ai(df, "Show 15 rows of data in tabular form columns Year, DIST_NAME and RICE_PRODUCTION_1000_tons DIST_NAME Bankura")
print(response)
```

Data structure

It is possible to get all the basic structures of a DataFrame with simple textual instructions.

```
response = pandas_ai(df, "Show structure of data ")
print(response)
```

The shape of the DataFrame can also be explored with this instruction.

```
response = pandas_ai(df, "what is the shape of data?")
print(response)
```

364 5

Data analysis

Overall basic statistics can easily be calculated with the following simple textual instruction:

```
response = pandas_ai(df, "what is the distribution of RICE_PRODUCTION_1000_tans?")
print(response)
```

```
count 364.000000
mean   641.068819
std    456.838868
min    51.490000
25%    289.392500
50%    493.310000
75%    848.247500
max    2145.290000
Name: RICE_PRODUCTION_1000_tons, dtype: float64
```

Skewness

To know the skewness of the distribution, you can use this simple query:

```
response = pandas_ai(df, "Show the skewness of RICE_PRODUCTION_1000_tons")
print(response)
1.2902333826377734
```

Graphics

Graphical representation is essential for effective data analysis. Distributions of different parameters